WORKED SOLUTION

STEP II 1996, Question 10

Pure • Circular Motion

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q10

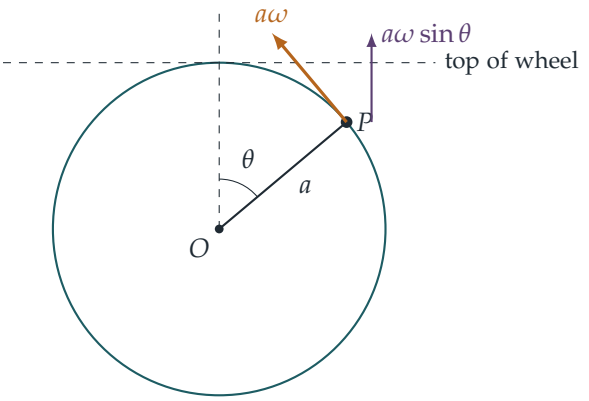
The key idea

At the moment she lets go, the heroine is a projectile whose initial position and velocity are fixed by where on the wheel she is. Only the vertical motion matters for the height, so the greatest height is her release height plus the extra height bought by the vertical component of her velocity. That gives the height as a function of a single release angle, and the question becomes: when can this function beat the top of the wheel, and what is its maximum?

Solution

Setting up

Model the heroine as a particle moving with the rim, and ignore air resistance. Let O be the centre of the wheel and suppose she lets go at the point P where OP makes an angle θ with the upward vertical, $0 \leq \theta \leq \pi$. Her speed on the rim is $a\omega$, directed along the tangent at P.



The tangent at P is perpendicular to OP, so it makes an angle θ with the horizontal. If

she is moving towards the top of the wheel when she lets go, her velocity has upward component $a\omega\sin\theta$ (shown in plum). If she is moving away from the top, her vertical velocity is downward or zero, so she never rises above her release point, which is no higher than the top; we may ignore those releases.

Her height above O at release is $a\cos\theta$. Under gravity alone her vertical velocity falls from $a\omega\sin\theta$ to zero, rising a further $(a\omega\sin\theta)^2/(2g)$, by $v^2 = u^2 - 2gs$. So her greatest height above the top of the wheel is

$$H(\theta) = a\cos\theta + \frac{a^2\omega^2\sin^2\theta}{2g} - a.$$

Write $c = \cos\theta \in [-1, 1]$ and $k = \dfrac{a\omega^2}{2g}$. Then $\sin^2\theta = 1 - c^2$ and

$$\begin{aligned}\frac{H}{a} &= c - 1 + k(1 - c^2)\\ &= (1 - c)[k(1 + c) - 1].\end{aligned}$$

Finding ω_0

At $c = 1$ (letting go at the top) $H = 0$, as we'd expect. For $c < 1$ the factor $1 - c$ is positive, so $H > 0$ exactly when $k(1 + c) > 1$.

> **Ask yourself** Over the allowed values of c, how large can $k(1 + c)$ be?

Since $c < 1$, $k(1 + c)$ takes every value in $[0, 2k)$, but never reaches $2k$. So:

- if $2k > 1$, we can choose $c < 1$ close enough to 1 that $k(1 + c) > 1$, and she rises above the top;
- if $2k \leq 1$, then $k(1 + c) < 2k \leq 1$ for every $c < 1$, so $H \leq 0$ for every release point and she can never rise above the top.

The condition $2k \leq 1$ is $a\omega^2 \leq g$. Note that the boundary case $\omega^2 = g/a$ belongs to the "not possible" side, so the largest such ω exists and is

$$\omega_0 = \sqrt{\frac{g}{a}}.$$

(This is also the angular speed at which a particle on top of the wheel would just lose contact if it were resting on the outside: the centripetal acceleration $a\omega^2$ equals g.)

The greatest height when $\omega > \omega_0$

Now $2k = a\omega^2/g = \omega^2/\omega_0^2 > 1$. As a function of c,

$$\frac{H}{a} = -kc^2 + c + k - 1$$

is a downward-opening quadratic, maximised where its derivative $1 - 2kc$ vanishes, at

$$c = \frac{1}{2k} = \frac{\omega_0^2}{\omega^2}.$$

This lies strictly between 0 and 1, so it is an allowed release point (a little before the top of the wheel), and it gives the maximum over all of $[-1, 1]$. Substituting,

$$\frac{H_{\max}}{a} = -\frac{1}{4k} + \frac{1}{2k} + k - 1 = k - 1 + \frac{1}{4k}.$$

With $k = \frac{1}{2}\omega^2/\omega_0^2$ and so $\frac{1}{4k} = \frac{1}{2}\omega_0^2/\omega^2$,

$$H_{\max} = \frac{a}{2}\left(\frac{\omega^2}{\omega_0^2} - 2 + \frac{\omega_0^2}{\omega^2}\right) = \frac{a}{2}\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)^2,$$

which is the required result. She achieves it by letting go where $\cos\theta = \omega_0^2/\omega^2$, while moving towards the top.

Checks

- At $\omega = \omega_0$ the formula gives 0, matching the boundary case; and it grows like $a\omega^2/(2\omega_0^2) = a^2\omega^2/(2g)$ for large ω, which is the height reached by a particle thrown straight up at speed $a\omega$ from the level of O. Indeed, for large ω the best release is near $\theta = \frac{\pi}{2}$, where the velocity is vertical.
- Units: ω/ω_0 is dimensionless, so $H_{\max}$ is a length.
- Numerically, with $a = 2\,\text{m}$ and $\omega = 2\omega_0$ the formula gives $\frac{a}{2}\left(2 - \frac{1}{2}\right)^2 = 2.25\,\text{m}$. Stepping the projectile from the release point $\cos\theta = \frac{1}{4}$ gives the same value, and a search over all release angles confirms that no other release does better.

Common slips

- Measuring the height from the centre of the wheel rather than from the top: the $-a$ in H matters.
- Assuming the best release is at $\theta = \frac{\pi}{2}$, where her velocity is vertical. That maximises the extra rise, not the total height.
- Forgetting to check that the optimal $c = \omega_0^2/\omega^2$ actually lies in $[-1, 1]$.

Beyond STEP

The method here, writing a quantity as a function of one control variable and optimising, is the simplest form of an optimal-control problem. A close cousin is the *envelope of trajectories* (the "parabola of safety"): the boundary of all points reachable from release

points on the wheel, which you can find by the same discriminant-style argument. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops projectile and rotational motion at first-year level.

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